

LensLab Studio

Selenium Test Suite

Automated End-to-End Quality Assurance

Python 3 · pytest · Selenium WebDriver · Chrome

Project Overview

What is it?

LensLab Studio is a web application for camera & lens rentals.

This test suite provides full end-to-end coverage of all critical user flows — from login through navigation and form submission to logout — using a real Chrome browser controlled by Selenium.

Technology Stack



Python 3.12+

Core programming language



pytest

Test framework & fixture management



Selenium WebDriver

Browser automation (Chrome)



Page Object Model

Architecture pattern for maintainability



WebDriverWait

Explicit waits — no fragile sleeps

Test Structure — AAA Pattern

1

SETUP

Load the page, authenticate if needed, navigate to the required section. Initialise fixtures and clear state.

2

ACTION

Simulate real user interactions: click buttons, type into fields, submit forms — exactly as a user would.

3

ASSERTION

Inspect the DOM and verify the expected outcome: element visible, error text matches, success shown.

✓ Tests are fully independent ✓ Failures are easy to diagnose ✓ Consistent, repeatable results

Key Test Cases

TestLogin

- Login page visible on startup
- Invalid credentials → error message shown
- Valid login → #app visible, login page hidden
- User initials 'AD' displayed in badge

TestNavigation

- Cameras page visible by default
- Navigate to Lenses / Contact / back
- Only ONE page visible at a time

TestContactForm

- Name required on submit
- Email: required / invalid format / valid clears error
- Phone: required / too short / valid clears error
- Successful submission → success message

TestLogout

- Logout → login page returns, #app hidden

Architecture — Page Object Model

Selectors

Centralised element locators (IDs, XPaths)

BasePage

find_element, click,
type_text, is_displayed

LoginPage AppPage ContactFormPage

High-level actions
e.g. login(), submit_form()

Fixtures

driver, valid_credentials,
valid_contact_data

Test Classes

TestLogin · TestNavigation
TestContactForm · TestLogout

✓ No duplicated selectors

✓ Tests read like plain English

✓ One change fixes all related tests

✓ Easy to extend with new pages

Running the Tests on Windows

1

Install Python 3.12+

```
python.org → check 'Add Python to PATH'
```

2

Create virtual environment

```
python -m venv venv
```

3

Activate environment

```
.\venv\Scripts\activate
```

4

Install dependencies

```
pip install selenium pytest webdriver-manager
```

5

Run the tests

```
pytest st2420018_py_test.py -v
```